LESSON PLAN

EWIS (21)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose and function of Altimeter and their types.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose and function of Altimeter and their types. Global positioning system (GPS) introduction, purpose airborne and space components of the system. Safety precautions to be observed whilst working on avionics systems
5. Specific Objectives	21.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Altimeter types and GPS components using L & Ds and IB; Instructor supervision required for safety precautions.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an altimeter?
(b) Name two types of altimeters used in aviation.
(c) Why is safety crucial when working on avionics systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic theory of communication • Purpose and function of Altimeter • Types of Altimeters • Global positioning system introduction • Airborne and space components of GPS	L & Ds	PPT, IB	15 min
(b)	• Safety precautions for avionics work • Static pressure system operation • GPS signal acquisition basics • Interpretation of altimeter settings • Maintenance requirements for altimeters	L & Ds	PPT, PDF	20 min
(c)	• Review of Altimeter types • GPS component functions • Safety protocol reinforcement • Troubleshooting common altimeter errors • Integration of GPS and Altimeter data	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between pressure altimeter and radar altimeter.
(b) Describe the purpose of the airborne component in the GPS system.
(c) List three safety precautions when handling avionics systems.

12. Home Assignment and Follow Up	Prepare a brief report on the next lesson topic: Purpose and function of radio systems including SLB and EL Emergency Radio System.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose and function of following radio systems: - SLB ELT Emergency Radio System	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Verify specific reference book titles for Altimeter theory in master syllabus.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min